QUANT BASED DATA SUFFICIENCY

Q1

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is A's speed?

I. A covers 10 km more per hour than B.
II. If B doubles his speed, then B covers 20 km per hour more than A does.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Your answer - ✓)

d) if the question cannot be answered even using I and II together.

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the ratio of the profit made by the two companies A and B?

I. The ratio of their monthly sales is 2 : 3.
II. The ratio of their monthly expenditures is 3 : 4.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Your answer - ✗)

d) if the question cannot be answered even using I and II together.

Solution:

Profit = monthly sales – monthly expenditures. As we do not have information about both monthly sales and monthly expenditures in any single statement, neither statement on its own is sufficient.
Combining both statements, let the monthly sales of A and B be ₹2x and ₹3x respectively. Let the monthly expenditures of A and B be ₹3y and ₹4y respectively.

$$\text{Ratio of profits} = \frac{2x-3y}{3x-4y} = \frac{2\frac{x}{y}-3}{3\frac{x}{y}-4}$$

As x/y is not known, this cannot be found. Choice (D)

> ***Directions for questions 1 to 15:*** Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

Did Nakul earn an interest which is equal to the principal?

I. Nakul deposited Rs.20,000 in a bank at 5% interest.
II. The amount was deposited for 20 years.

- **a)** if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.
- **b)** if the question can be answered using either statement alone.
- **c)** if the question can be answered using I and II together but not using I or II alone. (Your answer - ✕)
- **d)** if the question cannot be answered even using I and II together.

Solution:

From statement I, we do not know the time for which the amount was deposited. ∴ Statement I alone is not sufficient.
From statement II, we do not know the rate of interest.
∴ Statement II alone is not sufficient.
By combining I and II, we still do not know whether the interest is simple or compound.
Choice (D)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the value of b in the equation $x^2 - bx + c = 0$?

I. The roots are reciprocals of each other.
II. The sum of the roots is twice the product of the roots.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone.

d) if the question cannot be answered even using I and II together. (Your answer - ✕)

Solution:

From statement I, product of the roots = c = 1
But we cannot find the value of b.
∴ Statement I alone is not sufficient.
From statement II, b = 2c.
As we do not know the value of c, we cannot find b.
∴ Statement II is not sufficient.
By combining I and II, we get b = 2. Choice (C)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the percentage of maximum marks required to pass an exam?

I. Naveen got 30 percent of maximum marks and failed by 10 marks.
II. Sridhar got 75 marks and passed by 20 marks.

- **a)** if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.
- **b)** if the question can be answered using either statement alone.
- **c)** if the question can be answered using I and II together but not using I or II alone. (Your answer - ✅)
- **d)** if the question cannot be answered even using I and II together.

Solution:

Let x be the total marks in the exam.

From statement I, pass marks = $\frac{30x}{100} + 10$

$\therefore$ Percentage = $\dfrac{\frac{30x}{100} + 10}{x} \times 100$

But we do not know the value of x.
$\therefore$ Statement I alone is not sufficient.
From statement II, pass marks = 75 – 20 = 55
But we do not know the maximum marks. Hence percentage cannot be found.
$\therefore$ Statement II alone is not sufficient.

Combining both the statements, $\frac{30x}{100} + 10 = 55$

$\Rightarrow x = 150$

$\therefore$ Required percentage = $\frac{55}{150} \times 100$.

Choice (C)

What is the product of three single digit numbers a, b and c?

I. The LCM of a, b and c is 12.
II. The LCM of a and c is 4.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone. (Your answer - ✗)

c) if the question can be answered using I and II together but not using I or II alone.

d) if the question cannot be answered even using I and II together.

Solution:

Using statement I, a, b and c can be (1, 3 and 4) or (2, 3 and 4) or (3, 4 and 6) and so on. Hence it alone is not sufficient.
Using statement II alone (a, c) can be (1, 4), (4, 1), (2, 4) or (4, 2).
As b is unknown, statement II is not sufficient.
Combining both statements, if a = 1, b = 3 and c = 4, the product is 12. If a = 2, b = 3 and c = 4, the product is 24. The product is not unique. Choice (D)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the value of

$\frac{2y + x}{2y - x}$?

I. $3^{4x+y} = 243$
II. $2^{8y + x} = 512$

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Correct Answer - ✓)

d) if the question cannot be answered even using I and II together.

Solution:

From statement I we have $3^{4x+y} = 3^5$
i.e., $4x + y = 5$
We cannot find x and y.
From statement II, we have
$2^{8y+x} = 2^9$
i.e., $8y + x = 9$
We cannot find x and y.
Using both the statements we can find x and y.
Choice (C)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

Atmospheric air contains oxygen and carbon dioxide in the ratio 5 : 3 by weight. What is the ratio of the weights of oxygen and carbon dioxide in smog?

I. Exhaust gases contain oxygen and carbon dioxide in the ratio 1 : 11 by weight.
II. Smog is the mixture of atmospheric air and exhaust gases in the ratio 2 : 3 by weight.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Correct Answer - ✅)

d) if the question cannot be answered even using I and II together.

Solution:

The composition of atmospheric air (by weight) is O_2 and CO_2 in the ratio 5 : 3. We need the composition of smog. Clearly we need to consider only the combination of the two statements. Smog is defined in statement II and the composition of the exhaust gases, the term which occurs in statement II, is given in statement I.
We have the following

	Air	Exhaust Gases
O_2	5	1
CO_2	3	11
	8	12

or

Smog

Air	Exhaust Gases
2	3
8	12

We see that smog has 6 parts of O_2 and 14 parts of CO_2. Choice (C)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

Are a, b, and c in arithmetic progression?

I. 2a + b, 3b, 2c + b are in arithmetic progression.
II. a, b, a + c are in geometric progression.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone. (Correct Answer - ☑)

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone.

d) if the question cannot be answered even using I and II together.

Solution:

From statement I,
$2a + b - 3b = 3b - (2c + b)$
$2a - 2b = 2b - 2c$
i.e. $a - b = b - c$
Hence a, b, c are in AP.
Statement I alone is sufficient.
From statement II, a, b, a + c are in GP.
i.e. $b^2 = (a + c)a$
i.e. $b^2 = a^2 + ac$
$b^2 - a^2 = ac$.
So we can't say whether $a - b = b - c$ or not.
Hence statement II alone is insufficient. Choice (A)

> ***Directions for questions 1 to 15:*** Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the time taken by a train to cross a platform of 800 m?

I. The train crosses 400 m long platform in 12 sec.
II. The train crossed a moving person in 6 sec.

- **a)** if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.
- **b)** if the question can be answered using either statement alone.
- **c)** if the question can be answered using I and II together but not using I or II alone.
- **d)** if the question cannot be answered even using I and II together. (Correct Answer - ✅)

Solution:

Let the length and the speed of the train be x m and a m/s respectively.

We need to find $\frac{x+800}{a}$

From statement I, $\frac{x+400}{a} = 12$

From this we can't find $\frac{x+800}{a}$.

Statement I alone is not sufficient.

From statement II, we can't find the length or the speed of the train. So statement II alone is not sufficient.

Using both statements, we do not know the speed of the man and the direction of train with respect to the man.

So we can't find the time taken to cross 800 m long platform. Choice (D)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

How long was the director's speech?

I. The director spoke 35 words per minute.

II. Had the director spoke 7 fewer words per minute, his speech would have lasted 8 more minutes.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Correct Answer - ✅)

d) if the question cannot be answered even using I and II together.

Solution:

Using statement I, as we do not know how long the director spoke, we can not answer the question. Hence I alone is not sufficient.
Using statement II, let the number of words that the director spoke per minute be x, and the time be t minutes.
$\therefore xt = (x - 7)(t + 8)$.
Since we do not know x, t can not be found.
Therefore, II alone is also not sufficient.
Using I and II, from I we know x = 35.
Thereby substituting x in $xt = (x - 7)(t + 8)$, we can find t.
$\therefore$ I and II, taken together, are sufficient to answer the question. Choice (C)

What is the area of the triangle ABC?

∠

I. ABC = 30°

II. AC = 6 cm and BC = 12 cm.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

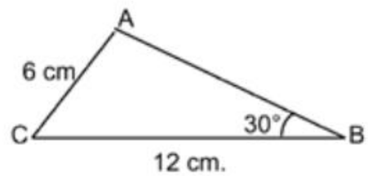
c) if the question can be answered using I and II together but not using I or II alone. (Correct Answer - ✅)

d) if the question cannot be answered even using I and II together.

Solution:

Statement I alone is not sufficient, as it gives no numerical data.
Statement II alone is not sufficient, as we don't know the information about the other side or any of the angles of the triangle.

From I and II, we have $\angle ABC = 30°$, AC = 6cm, BC = 12 cm



Applying Sine rule, we can see that $\angle A = 90°$ ($\because \frac{BC}{\text{Sin}A} = \frac{AC}{\text{Sin}B} \Rightarrow \sin A = 1$)

$\therefore$ The area of the triangle can be found. Choice (C)

> ***Directions for questions 1 to 15:*** Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the average monthly profit from sales of product A in a leap year?

I. In a month that has 30 days, the profit from the sales of product A is Rs.6000.
II. In a month that has 31 days, the profit from the sales of product A is Rs.6200.

- **a)** if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.
- **b)** if the question can be answered using either statement alone.
- **c)** if the question can be answered using I and II together but not using I or II alone.
- **d)** if the question cannot be answered even using I and II together. (Correct Answer - ✅)

Solution:

Neither statement alone is sufficient as each of them gives the information of the monthly profit only for months with 30 days or 31 days. Even by combining the two statements, we cannot find the answer as we don't know the monthly profit in the month with 29 days (February). Choice (D)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

An article is sold at 15% profit. What is the selling price of the article?

I. Had the article been sold for Rs.60 more, the profit would have doubled.
II. Profit made by selling the article is Rs.90.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone. (Correct Answer - ✅)

c) if the question can be answered using I and II together but not using I or II alone.

d) if the question cannot be answered even using I and II together.

Solution:

From statement I, if CP is ₹x, then SP = 1.15x. Given $(1.15x + 60) - x = 2[1.15x - x]$
Statement I alone is sufficient.
From statement II, 15% of CP = 90
$\therefore$ CP = ₹600.
Statement II alone is sufficient. Choice (B)

Directions for questions 1 to 15: Each question is followed by two statements, I and II. Indicate your responses based on the following directions:

What is the speed of the boat in still water?

I. The time taken by the boat to travel 20 km upstream is same as the time it takes to cover 30 km downstream.
II. The boat travels 36 km downstream in 6 hours.

a) if the question can be answered using one of the statements alone, but cannot be answered using the other statement alone.

b) if the question can be answered using either statement alone.

c) if the question can be answered using I and II together but not using I or II alone. (Correct Answer - ✅)

d) if the question cannot be answered even using I and II together.

Solution:

Let the speed of the boat in still water the x kmph. Let the speed of the water current be y kmph.

From statement I, $\frac{20}{x-y} = \frac{30}{x+y}$ ......(1)

We do not know the speed of the water. So we can't find the speed of the boat in still water.

Statement I alone is not sufficient.

From statement II, $x + y = \frac{36}{6} = 6$

Statement II alone is also not sufficient as we do not know the speed of the water current.

Using both statements, we have $x + y = 6$ and substituting this in (1), we get $x - y = 4$.

By solving them, we get the value of x. Choice (C)